

CL26 vs GG18 comparison (112-bit security)

Evaluation setup

Machine: Apple M2 Max, 12 cores, 32 GB RAM, macOS 26.5.2 (Darwin 25.5.0, arm64).

	CL26	GG18
Language	C++17	Go 1.23
Key dependencies	CMake build; BICYCL (bundled class-group library); GMP + GMPXX (find_library(... REQUIRED)); OpenSSL (find_package(OpenSSL REQUIRED))	github.com/bnb-chain/tss-lib/v3 v3.0.0 (module go.mod in adapters/tss-lib), plus its transitive deps: decred/dcrd/dcrec/secp256k1, btcsuite/btcd, gogo/protobuf, google.golang.org/protobuf, go.uber.org/zap, golang.org/x/crypto

Timing & throughput

n	t	Impl	Setup (ms)	DKG/Keygen (ms)	Sign (ms)	$\pm\sigma$ (ms)	Verify (ms)	Throughput (sig/s)
3	1	CL26	149.2	34.0	694.6	5.9	0.127	1.44
3	1	GG18	4,042.2	2,318.2	169.4	3.2	0.162	5.90
3	2	CL26	148.2	34.0	1,308.9	21.5	0.183	0.76
3	2	GG18	4,751.7	2,190.3	281.9	7.9	0.176	3.55
5	2	CL26	145.6	48.2	1,368.5	10.5	0.184	0.73
5	2	GG18	—	—	298.3	12.0	0.162	3.35
5	4	CL26	138.4	50.2	3,410.8	8.6	0.332	0.29
5	4	GG18	4,189.8	3,703.1	761.2	8.4	0.172	1.31
10	5	CL26	135.8	86.4	4,507.1	45.6	0.364	0.22
10	5	GG18	3,687.8	12,195.7	1,115.4	9.6	0.167	0.90

Bandwidth (one signing operation)

n	t	Impl	Total (B)	Per-party (B)
3	1	CL26	9,317	4,658
3	1	GG18	21,414	10,707
3	2	CL26	13,967	4,656
3	2	GG18	59,691	19,897
5	2	CL26	13,976	4,659
5	2	GG18	59,694	19,898

n	t	Impl	Total (B)	Per-party (B)
5	4	CL26	23,295	4,659
5	4	GG18	191,398	38,280
10	5	CL26	27,935	4,656
10	5	GG18	284,822	47,470

Object sizes (bytes)

n	t	Impl	Sig	EC PK	EC share	Enc PK	Enc share	Enc ctext
3	1	CL26	56	29	28	224	74	450
3	1	GG18	64	33	32	256	256	512
3	2	CL26	56	29	28	224	74	450
3	2	GG18	64	33	32	256	256	512
5	2	CL26	56	29	28	224	75	450
5	2	GG18	64	33	32	256	256	512
5	4	CL26	56	29	28	224	75	450
5	4	GG18	64	33	32	256	256	512
10	5	CL26	56	29	28	226	77	450
10	5	GG18	64	33	32	256	256	512

Correctness

n	t	Impl	All signatures valid
3	1	CL26	yes
3	1	GG18	yes
3	2	CL26	yes
3	2	GG18	yes
5	2	CL26	yes
5	2	GG18	yes
5	4	CL26	yes
5	4	GG18	yes
10	5	CL26	yes
10	5	GG18	yes

Acronyms

Acronym	Full form
CL26	Three-Round-Multiparty-ECDSA — a 3-round threshold-ECDSA protocol (also referred to in the schema as TR-ECDSA) built on CL-HSM class-group encryption
GG18	Gennaro–Goldfeder 2018 — the threshold-ECDSA protocol (9 signing rounds, Paillier encryption) implemented by bnb-chain/tss-lib
DKG	Distributed Key Generation
EC	Elliptic Curve
PK	Public Key
Sig	Signature
Enc	Encryption
ciphertext	Ciphertext